Lecture Scribe CSE 400

Tutorial_1_Solution.pdf
PDF

Example Lecture Scribe (Structure Only)

Definitions and Notation
Present the formal definitions and symbols introduced in the lecture.

Assumptions
List all assumptions explicitly stated during the lecture.

Theorems / Results
State all theorems and results exactly as covered in the lecture material.

Proofs / Derivations
Provide step-by-step derivations strictly following what is shown in the lecture.

Worked Examples
Include solved examples with intermediate steps only where they appear in the lecture.

Conclusion
Mention only the key outcomes discussed in the lecture.

You are provided with pdf of lecture material. Your task is to generate a lecture scribe suitable for exam reference, strictly following the structure above.

Use only the content from the provided slides and relevant textbook sections.

Include definitions, notation, assumptions, results, proofs, derivations, and examples only if they are explicitly taught in the lecture.

Show steps only up to the point demonstrated in the

lecture.

Maintain a formal academic tone and follow the lecture flow precisely.

Do not add explanations, intuition, summaries, new theory, or new examples.

Do not fill in missing steps.

Ignore informal content and images.

Do not infer or introduce any information that is not provided.

Lecture Scribe

CSE 400: Fundamentals of Probability in Computing
Tutorial–1: Solution
School of Engineering and Applied Science (SEAS), Ahmedabad University
February 13, 2026

Definitions and Notation

1. **Multinomial Formula**

$$\frac{n!}{n_1!n_2!\dots n_k!}$$

2. **Uniform Distribution**
   Arrival time uniformly distributed over an interval.
3. **Complement Rule**

$$P(A^c) = 1 - P(A)$$

4. **Inclusion–Exclusion Principle**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

5. **De Morgan's Law**

$$P(A^c \cap B^c) = 1 - P(A \cup B)$$

6. **Poisson Distribution**
   If $X \sim \text{Poisson}(\lambda)$,

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

7. **Gaussian (Normal) Distribution**

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right), \quad P(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

8. **Binomial Distribution**

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

9. **Cumulative Distribution Function (Discrete)**

$$F(a) = \sum_{x \leqslant a} p(x)$$

Assumptions

- Order within groups does not matter and groups are unlabeled (Q1).
- Arrival time is uniformly distributed over 30 minutes (Q2).
- Independence of events where stated (Q8, Q10).
- Poisson model applies for rare events in fixed intervals (Q4, Q5, Q6).
- Jurors act independently and each makes correct decision with probability θ (Q8).
- System components function independently with probability p (Q10).

Theorems / Results

1. **Multinomial Counting Result**

$$\frac{20!}{(5!)^4 4!}$$

2. **Poisson Complement Results**

$$P(X \geqslant k) = 1 - \sum_{i=0}^{k-1} P(X = i)$$

3. **Gaussian Symmetry**

$$\Phi(-x) = 1 - \Phi(x)$$

4. **Condition for Larger System Reliability**

$$p > \frac{1}{2}$$

Proofs / Derivations

Q1(a)

$$\frac{20!}{(5!)^4 4!}$$

Q1(b)

$$P = \frac{1}{\frac{20!}{(5!)^4 4!}} = \frac{(5!)^4 4!}{20!}$$

Q2

Total interval = 30 minutes.

(a)

Favorable time = 10 minutes.

$$P = \frac{10}{30} = \frac{1}{3}$$

(b)

Favorable time = 10 minutes.

$$P = \frac{10}{30} = \frac{1}{3}$$

Q3

Given:

$$P(L) = 0.15, \quad P(E) = 0.25, \quad P(L \cap E) = 0.08$$

$$P(E^c) = 1 - 0.25 = 0.75$$

$$P(L \cup E) = 0.15 + 0.25 - 0.08 = 0.32$$

$$P(L^c \cap E^c) = 1 - 0.32 = 0.68$$

$$P(L^c \mid E^c) = \frac{0.68}{0.75} \approx 0.907$$

Q4

$$\lambda = 0.2$$

(a)

$$P(0) = e^{-0.2} = 0.8187$$

(b)

$$P(1) = 0.2e^{-0.2} = 0.1637$$

$$P(X \geqslant 2) = 1 - 0.8187 - 0.1637 = 0.0176$$

Q5

$$\lambda = 3.5$$

$$P(0) = e^{-3.5} = 0.0302$$

$$P(1) = 3.5e^{-3.5} = 0.1057$$

(a)

$$P(X \geqslant 2) = 1 - 0.0302 - 0.1057 = 0.8641$$

(b)

$$P(X \leqslant 1) = 0.0302 + 0.1057 = 0.1359$$

Q6

$$\lambda = 3$$

$$P(0) = 0.0498$$

$$P(1) = 0.1494$$

$$P(2) = 0.2240$$

(a)

$$P(X \geqslant 3) = 1 - (0.0498 + 0.1494 + 0.2240) = 0.5768$$

(b)

$$P(X \geqslant 1) = 0.9502$$

$$P(X \geqslant 3 \mid X \geqslant 1) = \frac{0.5768}{0.9502} \approx 0.607$$

Q7

Given:

$$m = -3, \quad \sigma = 2$$

1.

$$P(X \leqslant 0) = \Phi\left(\frac{3}{2}\right) = 1 - Q\left(\frac{3}{2}\right)$$

2.

$$P(X > 4) = Q\left(\frac{7}{2}\right)$$

3.

$$P(\mid X + 3 \mid < 2) = 2\Phi(1) - 1 = 1 - 2Q(1)$$

4.

$$P(\mid X - 2 \mid > 1) = 1 - Q(2) + Q(3)$$

Q8

If innocent:

$$\sum_{i=5}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i}$$

If guilty:

$$\sum_{i=8}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i}$$

Overall:

$$\alpha \sum_{i=8}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i} + (1-\alpha) \sum_{i=5}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i}$$

Q9

$$ce^{\lambda} = 1 \Rightarrow c = e^{-\lambda}$$

(a)

$$P(0) = e^{-\lambda}$$

(b)

$$P(X > 2) = 1 - e^{-\lambda} - \lambda e^{-\lambda} - \frac{\lambda^2 e^{-\lambda}}{2}$$

CDF step function example as provided.

Q10

(a)

5-component effective:

$$10p^3(1-p)^2 + 5p^4(1-p) + p^5$$

3-component effective:

$$3p^2(1-p) + p^3$$

Condition reduces to:

$$3(p-1)^2(2p-1) > 0$$

$$p > \frac{1}{2}$$

(b)

$$P_{2k+1} - P_{2k-1} = \binom{2k-1}{k} p^k (1-p)^k (2p-1)$$

$$> 0 \;\Leftrightarrow\; p > \frac{1}{2}$$

Worked Examples

- Uniform waiting time computation (Q2).
- Poisson probability computations (Q4–Q6).
- Gaussian Q-function conversions (Q7).
- Jury conditioning computation (Q8).
- Discrete CDF step function example (Q9).
- Binomial system reliability comparison (Q10).

Conclusion

- Multinomial counting applied for equal partitioning.
- Complement rule used for Poisson tail probabilities.
- Conditional probability applied using complements and inclusion–exclusion.
- Gaussian probabilities expressed using Φ and Q functions.
- Larger odd-component systems are better when $p > \frac{1}{2}$.